

What (if any) consequences do Gödel's incompleteness theorem results have for our understanding of the relationships between human minds, mathematics and machine computation?

In this paper, I will argue that there do exist several consequences regarding minds, math, and machine computation due to Gödel's incompleteness theorem results. Starting at the base of logic, these results have repercussions in mathematics and computations and lead us to wonder many things about our own existence.

We first start with Hilbert who believed, as did many mathematicians of his time, that every mathematical truth is provable, i.e. it has a proof. He posed this within the second of his 23 unsolvable problems at the beginning of the 20th century: show that all of mathematics is consistent. Consistency of a formal system is defined as “the property that ‘there is no formula ϕ such that both ϕ and $\neg\phi$ are provable’”¹.

Soon after the problem was posed, Gödel showed that for every mathematical axiomatic system, whose formulae can be procedurally generated (i.e. via an algorithm), is not capable of proving all truths within the system. This is known as Gödel's First Incompleteness Theorem (GT1). His Second Incompleteness Theorem (GT2) states that any system as in GT1 cannot prove its own consistency.

Starting with GT1, Gödel created a way to encode every well-formed formula (wff) within a formal system by way of Gödel numbering. An example of this type of numbering is to convert English into a sequence of decimal numbers by using ASCII. Since a prime number is only factorable by 1 and itself, Gödel used the property of prime factorization for his sequence of numbers: every natural number can be uniquely expressed as a product of primes. Given a wff, assign a unique number to every symbol, producing a sequence of n numbers and take the first n primes to the power of this sequence. For example, given the wff “ $1 = 1$ ” assign the following Gödel numbers (‘1’: 6, ‘=’: 5) and then do the encoding: $2^6 \bullet 3^5 \bullet 5^6 = 243 \bullet 10^6$. It can also be shown that we can apply Gödel encoding to a sequence of wffs, though I will not show that in this paper. Next, he created a function which takes two Gödel numbers, n and m , and returns true if the sequence of formulae represented by n constitutes a proof for the formula represented by m . By this function, we can check if any formula is provable by the axioms of a given system. Using this encoding method, Gödel encoded the formula “I am not provable from the axioms”.

The above formula can be provably shown to neither provable nor disprovable within the system. If we call the formula S and assume it to be false, then it must be provable from the axioms. If S is provable, then it is true. Since it can not be both true and false, by consistency of the system, our original assumption was wrong, and it is indeed true. Thus, we have a true formula that cannot be proved from the axioms. This is known as a Gödel sentence, denoted G_F , for a system F . If we decide to include G_F as an axiom of a new system F' to bypass this issue, we can show using the same procedure that F' also has its own Gödel sentence $G_{F'}$, and so on recursively.

While GT1 is a very elaborate and interesting proof, I find GT2 to be a much more interesting proof. I won't go into the actual proof since it is quite complicated but the sketch is as follows. Let the consistency of F be denoted by $\text{Cons}(F)$, the formula that states “there is no natural number that codes a derivation of a contradiction from the axioms of F ”. It is possible to show that $\text{Cons}(F)$ is not provable within F , i.e. $\text{Cons}(F)$ is a Gödel sentence for F . Thus, since every system has a Gödel sentence, every system cannot prove its own consistency.

A very interesting by-product of the way that GT1 is constructed is that you cannot simply construct a set of axioms that would render the theorem unprovable and then use those to do arithmetic. This is since the proof itself is based on arithmetic and as such requires an appropriate axiomatic system. Thus, there is no way for us to evade GT1. Now that I've covered the theorems themselves, let's move on to the implications.

First off, the field of artificial intelligence has gained lots of traction in recent years. The goal of this field is to recreate the human brain using computer programs. In his 1959 paper, *Minds, Machines, Gödel*, J.R. Lucas argues that this isn't possible. He states that an algorithmic automaton, also known as a Turing machine, that models

a human mathematician is an impossibility. A Turing machine is a “mathematical model of computation that defines an abstract machine which manipulates symbols on a strip of tape according to a table of rules”.⁶ This “table of rules” is a set of commands that are pre-programmed onto our Turing machine and can be treated as a set of axioms. Since we allow our Turing machine to perform mathematics, which includes basic arithmetic, this is a formal system. As such, by GT1, there must exist some true formula which is unprovable in this system, i.e. unprovable by the Turing machine, but is evidently true from outside the system, i.e. by a human mathematician.

This leads me to one of Kant’s beliefs: computer aided math (CAM) is not valid math. He believed that mathematical thought was in the mind and that CAM was simply an extension of empirical science. His argument was that since a computer assisted proof (CAP) could not be proved by a human mind, we have no way of verifying its certainty. Kant had issues more so with uncertainty and less with the computing correctness of the CAP, as we cannot know if a random bit had been flipped during a computation, leading to a false result. However, by Lucas’s argument, I believe that it is possible that CAM can be detrimental to humans. Since there exists formulae for which a Turing machine cannot provide a solution, but for which humans can evidently see the solution (but are not capable to prove since it might be too large of a proof), we have no way of verifiably proving the formulae.

This is similar, in some ways, to arguments raised by Paul Benacerraf and Douglas Hofstadter. They argue against Lucas’ ideas saying that it is not necessarily provable that this Turing machine to simulate human brains is impossible, rather that there exists a disjunction (Benacerraf’s view) or that there is an impasse between the machine and the brain (Hofstadter’s view). We already know that we cannot escape the effects of GT1 by using recursion, that is we cannot simply add the Gödel sentence of F , G_F , as an axiom of F' , since F' will itself have a Gödel sentence $G_{F'}$. In the same way, we cannot add G_C , the Gödel sentence for a computer system C , as an axiom since there will exist $G_{C'}$. However, Hofstadter argues that if we complete this process of adding the Gödel sentence to a system as an axiom, we increase the complexity of the system with every recursion. Eventually, our system arrives to an incredibly complex point where its Gödel sentence, denoted now by G_{C^n} , cannot be determined by the humans outside of the system. We denote this incredibly complex system $C^{'''''} = C^n$. By virtue of being the Gödel sentence for that system, the system also cannot produce G_{C^n} . Thus, the machine and the human that it models are equal in power.

Benacerraf’s view is similar in that he assumes right away that a formal system needed to model the entirety of the human brain, denoted again C^n , must be immensely complex if it even exists. He states that in order to be able to produce a Gödel sentence for a formal system, one is required to have a very good understanding of a system’s axioms and how they interact and behave. In fact, he says that the complexity of C^n must be so high that the human mind would not have a profound enough understanding of it to be able to produce G_{C^n} . Thus, either there is no formal system C^n that models the human adequately or the system is so complex that its subject cannot even understand it.

One of the big advantages of humans over a computer program is our consciousness. We are able to freely move about different systems, entering and exit them at will, in order to better understand everything around us. This is what allows us, for example, to enter an axiomatic formal system sufficient for arithmetic and understand why the formula “I am unprovable by the axioms of this system” makes sense. We can see the effects of the formula and then step back and exit the system to ponder the system itself. This meta property is what distinguishes us from being defined solely by a formal system. To translate this meta property to an axiomatic system, it would have to be a system, let’s say C^n , that can switch between different versions of itself. For example, the system would have to be able to switch to $C^{''''}$, or $C^{''}$, or even to C and then switch back to C^n and reflect on those other systems it converted to.

In terms of AI, we have not arrived at such a system yet since this would require code that can edit its own source files. However, to be able to do so effectively, it would have to know everything about itself in order to retain the possibility of returning to its original state after entering and exiting another state. This presents an issue as there are large gaping holes in humanity’s knowledge of itself, both physically and metaphysically. Thus, I believe that due to GT1 and GT2, and other theories derived from them, we are not able to effectively state whether or not we could construct an algorithmic automaton that could completely model the human mind. In that sense, I believe that this problem is a Gödel sentence, of sorts, for the system that we are currently in and we need to make further advances in order to determine whether or not we can provably provide an answer.

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